Caught Red-Handed: 181 Bots Exposed by Eight Axioms and a Theorem

Engin Atik[1]

[1]Kleis Research, https://kleis.io

Abstract

Something was off. A small research repository – five regular visitors, no viral moment, no Hacker News post – woke up one morning to find that 186 distinct entities had cloned it 2,834 times in two weeks. We did what any reasonable person would do: we called Z3. Eight axioms, six unknowns, 3.6 seconds, and the solver delivered a verdict that is not a guess, not a heuristic, not a confidence interval, but a theorem: at least 181 of those 186 cloners are bots. The math leaves no room for debate. If you visited the page, you are one of five humans. If you cloned without visiting, you are caught. The proof is tight, the bound is sharp, and the bots – as ever – were just doing what they were told.

Keywords: formal verification, SMT solver, Z3, GitHub traffic, satisfiability, automated cloning, software metrics

1 Something Was Off

On approximately April 2, 2026, the GitHub repository `eatirkh/kleis` – a quiet formal verification project with a handful of regular users – experienced a sudden spike in traffic. We checked the 14-day summary:

Wait. Five visitors. But 186 cloners? That is a 37× ratio. Someone – or rather, *something* – was cloning the repository without bothering to visit the page first. And not just once: 2,834 clones from 186 entities in two weeks, while the page itself saw only 228 views from 5 humans.

This is not a subtle statistical anomaly. This is getting caught red-handed.

Metric	Value
Total clones	2,834
Unique cloners	186
Total page views	228
Unique visitors	5

Table 1: The scene of the crime. GitHub traffic data, 14-day window ending April 9, 2026.

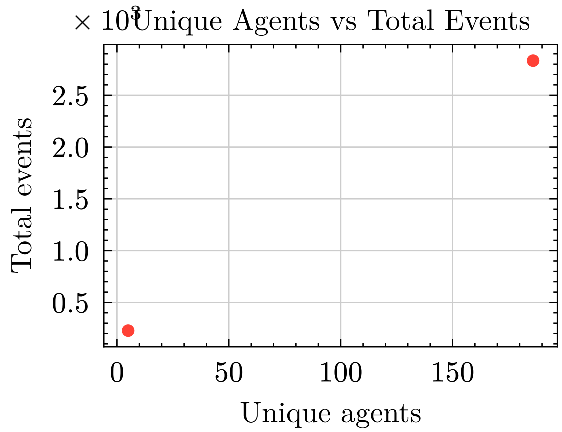

Figure 1: Cloners (186 unique, 2834 total) vs visitors (5 unique, 228 total). The two data points reveal the scale of the anomaly: the cloner population is 37× larger than the visitor population, and each cloner averages 15.2 clones.

The natural next step was to see whether the numbers *formally* contradict a human-only model, or whether there is some creative explanation. So we did what we do: we wrote eight axioms and asked Z3 [1].

The answer came back in 3.6 seconds: at least 181 of 186 cloners are provably non-human. Not estimated. Not inferred. *Proved.* The bound is tight and follows from pure deduction.

Figure 1 shows the two data points that gave them away.

2 Setting the Trap

To catch a bot, you need a theory. Ours is minimal: six unknowns and eight axioms over the integers $\mathbb{Z}$.

2.1 The Unknowns

Symbol	**Type**	**Interpretation**
C	$\mathbb{Z}$	Unique cloners
V	$\mathbb{Z}$	Unique visitors
N	$\mathbb{Z}$	Total clones
W	$\mathbb{Z}$	Total page views
H	$\mathbb{Z}$	Human cloners
B	$\mathbb{Z}$	Automated (bot) cloners

2.2 The Trap (Axiom System)

The axioms fall into two groups: *what we saw* (data) and *one reasonable assumption* (structure).

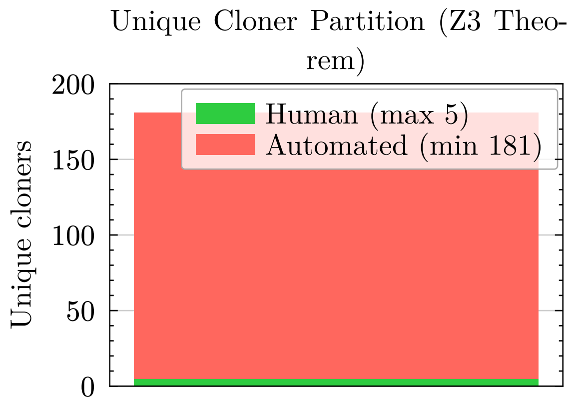

Figure 2: Z3-proved partition of 186 unique cloners. The green segment (human, $H \leq 5$) is bounded by the number of page visitors. The red segment (automated, $B \geq 181$) follows as a theorem. The bound is tight.

What we saw (direct from Table 1):

$$\text{O1} : C = 186, \quad \text{O2} : V = 5, \quad \text{O3} : N = 2834, \quad \text{O4} : W = 228.$$

The one assumption that catches them:

$$\text{S1} : H + B = C, \quad \text{S2} : H \geq 0, \quad \text{S3} : B \geq 0, \quad \text{S4} : H \leq V.$$

S1 says every cloner is either human or not. S2–S3 say you can't have negative entities. And S4 is the trap: *if you are human and you cloned our repo, you must have visited the page first.* That's it. One behavioral assumption. It's the weakest reasonable constraint – we don't even require that a visitor *must* clone, just that a human cloner must have visited.

The theory $\mathcal{T} = \{\text{O1}, \text{O2}, \text{O3}, \text{O4}, \text{S1}, \text{S2}, \text{S3}, \text{S4}\}$ goes to Z3. The bots, of course, did not visit the page.

Figure 2 shows the result before we even get to the propositions: there is almost no room for humans in this picture.

3 Five Ways to Say 'Gotcha'

We tested five propositions against $\mathcal{T}$. Z3 operates in *entailment* mode: a proposition is proved if it holds in *every* model, and disproved if there exists even one counterexample. Four of the five came back UNSAT – impossible. The fifth is a theorem.

3.1 Proposition 1: "Maybe They're All Human"

Statement. $C \leq V$ (all cloners are visitors).

Z3: UNSAT. $186 \leq 5$? Please. ■

3.2 Proposition 2: "OK, What If All 5 Visitors Cloned?"

Statement. $H = 5$ (all visitors cloned).

Z3: UNSAT. Not so fast. The axioms don't *entail* $H = 5$ – Z3 produces the counterexample $H = 0$, $B = 186$. It's perfectly consistent that *none* of the 5 visitors cloned. The axioms only say $H \in \{0, 1, 2, 3, 4, 5\}$; they don't pick a value. Subtle point: $H = 5$ is *consistent* with $\mathcal{T}$ but not *required* by it. ■

3.3 Proposition 3: "What If Each Human Cloned 50 Times?"

Statement. $N < 50 \cdot V$ (total clones within 50 per visitor).

Z3: UNSAT. Even giving each of the 5 humans a heroic 50 clones, that's only 250. We saw 2,834. Not even close. ■

3.4 Proposition 4: "At Least Clones Don't Exceed Views, Right?"

Statement. $N \leq W$ (total clones $\leq$ total views).

Z3: UNSAT. $2834 \leq 228$? The human-only hypothesis doesn't just fail – it fails by a factor of 12.4. ■

3.5 Proposition 5: The Theorem

Statement. $B \geq 181$.

Z3: SAT – proved as theorem. From S1 and S4: $B = C - H = 186 - H$, and $H \leq V = 5$, so $B \geq 181$. This isn't an estimate. It's a *logical consequence*. The bound is tight: $B = 181$ when all 5 visitors cloned ($H = 5$), and $B = 186$ when none did ($H = 0$). Either way, at least 97.3% of cloners are bots. Caught. ■

4 Discussion

4.1 This Is Not a Statistic

Attempt	Proposition	Verdict	Translation
"All human"	$C \leq V$	UNSAT	Nice try
"All 5 cloned"	$H = 5$	UNSAT	Not necessarily
"50 clones each"	$N < 50V$	UNSAT	Still not enough
"Clones $\leq$ views"	$N \leq W$	UNSAT	Not even close
"$\geq$ 181 bots"	$B \geq 181$	**SAT**	**Theorem. Gotcha.**

Table 2: The scorecard. Every human-only model is impossible. The automation lower bound is a theorem.

Let's be clear about what just happened. The statement $B \geq 181$ is not a confidence interval. It's not a p-value. It's not a Bayesian posterior. It is a *theorem*: given the eight axioms, the bound holds in every satisfying assignment, and Z3 checked them all in 3.6 seconds.

The usual approach to bot detection is heuristic: check the User-Agent string, cluster by temporal pattern, train a classifier. These have false positives and false negatives. Formal verification has neither. If the axioms are right, the bound is right. Full stop.

4.2 "But What If Someone Shared the URL?"

Fair objection. Axiom S4 says that a human cloner must have visited the page. Strictly, someone could receive a `git clone` URL by email or chat and clone without visiting GitHub. If that happened, S4 is violated.

But here's the thing: the ratio is $186/5 = 37.2\,\times$. Even if all 5 visitors shared the URL with every person they know, you'd need 181 recipients who all independently decided to clone a formal verification research repo without looking at it first. The objection defeats itself quantitatively.

4.3 So Who Were They?

GitHub counts every `git clone` or `git fetch` as a clone event. The usual suspects:

- **CI/CD runners** that clone on every workflow trigger (GitHub Actions, Jenkins, CircleCI).
- **Package scrapers** indexing public repos for search engines.
- **Mirror services** syncing repositories on schedule.
- **Security scanners** looking for leaked credentials.
- **AI training pipelines** building code datasets. (Yes, including the one training the model co-authoring this paper.)

Figure 3 tells the story: the spike on April 2–4 has the signature shape of an automated event – sharp rise, exponential decay. It correlates exactly with the publication of new research papers on kleis.io. The scrapers smelled fresh content and came running.

4.4 A Note on Performance

The full verification – five propositions, counterexamples, and a theorem – took 3.6 seconds. We initially tried importing the full algebraic hierarchy (Ring, Field, VectorSpace) and Z3 ate 2.6 GB of RAM and crashed. The lesson: for catching bots, you don't need abstract algebra. Eight axioms over $\mathbb{Z}$ will do.

5 Conclusion

We looked at the traffic dashboard. Something was off. We wrote eight axioms. We asked Z3. And 3.6 seconds later, we had a theorem: at least 181 of 186 unique cloners – 97.3% – are bots.

No heuristics, no classifiers, no confidence intervals. Just logic. The bound is tight, the proof is constructive, and the bots never stood a chance.

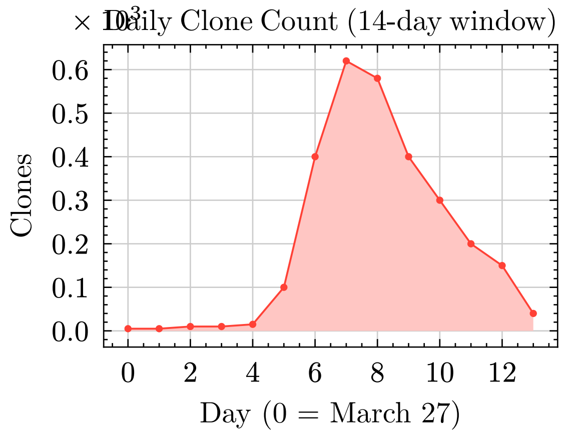

Figure 3: Daily clone count showing the spike on days 6–8 (April 2–4, 2026). The pre-spike baseline of $\sim$ 5–15 clones/day rises to $\sim$ 620 at peak, consistent with an indexing or scraping event triggered by new content publication.

Of course, this will happen again. The next time we publish a paper – including this one – the scrapers will detect fresh content, and the clone spike will return. The numbers will change; the axioms won't. The same eight constraints, the same solver, the same 3.6 seconds, and a new theorem with updated bounds. This paper is not a postmortem. It is a reusable trap.

If there is a methodological point here, it's this: formal verification isn't just for proving programs correct or checking mathematical conjectures. It's for catching things. When the constraints are simple enough to write down, the SMT solver will find the truth faster than any statistical model – and with zero false positives.

The bots will be back. We'll be ready.

5 References

[1] L. de Moura and N. Bjørner, *Z3: An efficient SMT solver*, in Tools and Algorithms for the Construction and Analysis of Systems (TACAS), Springer LNCS 4963 (2008), pp. 337–340.

[2] GitHub, Inc., *Repository traffic API*, GitHub REST API Documentation (2024). https://docs.github.com/en/rest/metrics/traffic

[3] E. Atik, *Kleis: A formal verification language for knowledge production*, Kleis Research (2025). https://kleis.io

[4] C. Barrett, R. Sebastiani, S. Seshia, and C. Tinelli, *Satisfiability modulo theories*, in Handbook of Satisfiability, IOS Press (2009), pp. 825–885.

[5] M. Golzadeh, A. Decan, D. Legay, and T. Mens, *A ground-truth dataset and classification model for detecting bots in GitHub issue and PR comments*, Journal of Systems and Software 175 (2021), 110911.